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Internal energy

Particle model of matter
Paper 1 · Combined and Separate
12 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Define internal energy.**	The total energy stored by the particles that make up a system.	☐
2	**State how internal energy is related to the energies of the particles.**	Internal energy = total kinetic energy + total potential energy of the particles	☐
3 ★	**State which part of the internal energy changes when the temperature changes.**	The kinetic energy of the particles	☐
4	**Describe how the average kinetic energy of the particles is related to temperature.**	The greater the average kinetic energy, the higher the temperature.	☐
5 ★	**State which part of the internal energy changes when a substance changes state.**	The potential energy of the particles	☐
6	**The potential energy of the particles changes at a constant temperature. State what happens to the substance.**	It changes state.	☐
7	**State the two things that can happen when a substance is heated.**	Its temperature can increase, or it can change state.	☐
8	**Describe the change in temperature of a substance during a change of state.**	The temperature remains constant.	☐
9	**State the two things that can happen when a substance is cooled.**	Its temperature can decrease, or it can change state.	☐
10	**In which state of matter is the internal energy of a substance highest?**	Gas	☐
11	**State what heating does to the internal energy of a system.**	It increases it (the particles gain energy).	☐
12	**State three things that the temperature rise of a heated substance depends on.**	Its mass, its specific heat capacity and the energy transferred to it	☐

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